Experiment: Creation of a Virtual Hard Disk (VHD)

Allocating Storage using Oracle VirtualBox

AIM

To create a Virtual Hard Disk (VHD) and allocate storage space to it using Oracle VirtualBox, so that it can be used as the storage medium for a virtual machine.

THEORY

A Virtual Hard Disk is a file stored on the host machine's physical disk that emulates a real hard drive for a virtual machine (VM). VirtualBox supports several disk image formats such as VDI (VirtualBox Disk Image), VMDK (Virtual Machine Disk), VHD (Virtual Hard Disk), HDD, and QED. Storage on a virtual disk can be allocated in two ways: **Dynamically Allocated**, where the file grows in size as data is added up to the maximum limit, and **Fixed Size**, where the full storage space is reserved on the host disk at the time of creation, offering faster read/write performance at the cost of using more disk space upfront.

PROCEDURE

1. Install and open **Oracle VirtualBox Manager** on the host machine.
2. Click on the **New** button on the toolbar to start creating a new virtual machine, or select an existing VM and go to **Settings > Storage** to add a new disk.
3. In the **Create Virtual Machine** wizard, enter a **Name** for the VM, choose the **Machine Folder**, and select the **Type** and **Version** of the guest operating system to be installed, then click **Next**.
4. In the **Hardware** step, allocate the required **base memory (RAM)** for the virtual machine and click **Next**.
5. On the **Virtual Hard Disk** screen, select the option **Create a Virtual Hard Disk Now** and click **Create**.
6. In the **Hard Disk File Type** window, choose the disk image format (e.g., **VDI - VirtualBox Disk Image**) and click **Next**.
7. In the **Storage on Physical Hard Disk** window, choose between:
   (a) **Dynamically Allocated** — disk file grows as needed, up to the set limit.
   (b) **Fixed Size** — full disk space is reserved immediately.
   Click **Next** after selecting the appropriate option.
8. In the **File Location and Size** window, specify the **file name/location** for the virtual disk and use the slider or input box to set the desired **storage size** (e.g., 20 GB, 50 GB), then click **Create**.
9. VirtualBox allocates the storage and creates the .vdi (or selected format) disk file at the specified location. The new virtual machine now appears in the VirtualBox Manager list with the allocated virtual hard disk attached.
10. Verify the disk by opening **File > Virtual Media Manager** in VirtualBox, where the newly created VHD is listed along with its size, type, and attached VM.
11. (Optional) Start the VM and proceed to install the guest operating system, which will use the newly created virtual hard disk for storage.

Sample Configuration Used

Parameter	Value
Disk Format	VDI (VirtualBox Disk Image)
Allocation Type	Dynamically Allocated
Storage Size	20 GB
File Location	Default VirtualBox VM folder
Guest OS Type	As selected during VM creation

RESULT

A Virtual Hard Disk of the specified size (e.g., 20 GB, VDI format, dynamically allocated) was successfully created and attached to the virtual machine using Oracle VirtualBox. The virtual disk file was generated at the chosen location on the host system, and its presence was confirmed through the VirtualBox Media Manager. Thus, the storage was successfully allocated for use by the virtual machine.

Conclusion: The experiment of creating a Virtual Hard Disk and allocating storage using VirtualBox was performed successfully.